

Futexes and Hybrid Locks

Spinlocks vs. sleeping locks: System calls are expensive: going to sleep is expensive. Spinning involve doing work all the time: being awake is expensive.

Spinlock: Threads intensively check whether lock is free. Checks performed entirely in user space: no system calls required. Great when there is low contention, or short critical sections. Prevents CPU from doing other useful work when contention is high.

Sleeping lock: Thread that doesn't manage to get lock asks OS to put it to sleep. OS wakes thread up when lock becomes free: requires system calls. Good idea to sleep if lock won't be free for a while: CPU can do other things. Bad idea to sleep if lock is just about to become free.

Futex: fast userspace mutex: futex is a Linux system call - exposes functionality offered by the kernel (so not userspace). But futex works on userspace data, and can be used to implement synchronisation primitives that execute primarily in userspace. futex is very flexible and can implement more than just mutexes.